Manifold-Aware Concept Erasure

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Abstract

Concept erasure aims to remove target information from a representation without destroying other information it carries. This is difficult for nonlinear concepts: removing the signal used by one probe does not prevent a fresh nonlinear probe from recovering the concept, while unconstrained gradient-based edits can move representations away from the support of natural inputs. We introduce TanGCE, a manifold-aware erasure method that constrains nonlinear edits to local tangent directions. For each representation, TanGCE estimates a tangent basis from nearest-neighbor secants, projects the concept-scorer gradient onto this basis, and scales the resulting step by the local singular spectrum, so motion along each tangent axis is proportional to the neighborhood's extent along that axis. Before the iterative tangent loop, we remove dominant low-order target signal with closed-form LEACE and a rank-2 covariance-asymmetry projection. Across 119 panels, covering 13 language models, three NLP concepts, and 40 CelebA-CLIP attributes under two control regimes, TanGCE uses one fixed set of hyperparameters with no per-panel tuning. On the 39 NLP panels, it reaches the majority-class probe-accuracy floor on 26–33 panels across control-damage budgets, compared with 22–26 for Obliviator, while preserving control accuracy better. On CelebA-CLIP, it is the only method that retains an admissible operating point for every attribute in the highly correlated control regime.

1 Introduction

Neural representations encode many attributes of their inputs: semantic content, style, demographic information, task labels, and other latent factors. In many applications, we may want to remove one such attribute, for example a protected or spurious concept, while preserving everything else the representation contains [De-Arteaga et al., 2019, Avitan et al., 2025]. This is the central tension in *concept erasure*: removing the target information while unnecessarily damaging other control concepts, known as *surgicality* (Yanai:) is it actually known? have previous papers used this term? mayes!.

The surgicality requirement makes erasure especially difficult. A target concept is rarely encoded in isolation. It may be entangled with many other factors, most of which are unlabeled and therefore cannot be explicitly protected. Moreover, erasure must address *recoverability*: even when a chosen classifier is driven to chance, a freshly trained classifier on the edited representations may still recover the protected attribute [Elazar and Goldberg, 2018]. Linear erasure methods provide simple and effective closed-form interventions [Ravfogel et al., 2020, Belrose et al., 2023], but they only remove linearly accessible information and do not generally block nonlinear probes [Ravfogel et al., 2022]. Nonlinear erasure, however, introduces a different danger: if we directly follow the gradient of a nonlinear concept scorer in the ambient representation space, the edit may remove target signal by moving the representation to a region that natural inputs never occupy.

While neural representations live in $\mathbb{R}^d$, learned representations that are produced by different models do not fill this space uniformly. Instead, they concentrate near a much smaller, structured support. This observation changes the geometry of erasure; an arbitrary direction in the ambient space may be a valid vector in $\mathbb{R}^d$, but it need not be a valid direction along the representation support. Moving in

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such a direction can push the representation off-support, distorting unrelated information and harming control concepts. We therefore ask whether nonlinear (Yanai:) why is this related to non linearity? erasure can be made more surgical by forcing the edit to remain locally compatible with the geometry of natural representations.

We introduce TanGCE, a tangent gradient-based method for nonlinear concept erasure. Around each representation, we estimate a local tangent space using the SVD of nearest-neighbor secants. Instead of applying the ambient gradient (Yanai:) what's that? of a concept scorer, we project that gradient onto the estimated tangent space and edit only along directions supported by the local neighborhood. This constrains erasure to directions that are locally plausible under the *representation manifold*, reducing the risk of off-support interventions.

A second ingredient is needed for stability. The local tangent space is not isotropic: different tangent directions correspond to different local scales of variation. A step of the same Euclidean length can be harmless along a high-variance direction but destructive along a low-variance one. We therefore use the singular spectrum produced by the same local SVD to precondition the tangent step. After projecting the scorer gradient onto the tangent space, we express it in the local principal-coordinate system and rescale each coordinate by its corresponding local scale. This local-σ preconditioner ties the magnitude of the intervention in each direction to how far the representation neighborhood actually extends in that direction, yielding a per-row trust region that transfers across models and layers.

Our full pipeline has two stages. First, we remove dominant low-order target signal in closed form. We apply LEACE to remove the first-order mean-difference direction between the two target classes [Belrose et al., 2023], and then remove a rank-2 subspace corresponding to the leading covariance-asymmetry directions of the class-conditional covariance difference. This sacrifices only a small number of representation dimensions while reducing the strongest linear and quadratic traces of the target concept. Second, we run tangent-constrained nonlinear erasure on the remaining signal using the local-σ preconditioned tangent gradients described above.

(Yanai:) the entire method is described across three paragraphs - that's too much. make it more concise

The empirical signature of this design is surgicality (Yanai:) strange phrasing. We evaluate TanGCE on 119 panels (Yanai:) idk what panels are (well, you told me, but that's not a common term). experiments/setups? under fixed pipeline hyperparameters, without per-panel tuning: 13 language models $\times$ 3 concepts at 50%-depth, and CelebA-CLIP $\times$ 40 attributes $\times$ 2 control-set regimes. At a fixed control-damage budget of $\Delta Y \geq -5$pp, TanGCE reaches the per-panel majority-class probe-accuracy floor on 33 of 39 NLP panels, compared with $22/39$ for Obliviator, while damaging the control task less; the same pattern holds at every other budget we test ($\leq$ 1pp, $\leq$ 3pp, $\leq$ 10pp). (Yanai:) you use many terms here w/o explaining them (panel, control-set regimes, damage budget, NLP panel, Obliviator). re-write this paragraph. focus on the high level idea: we remove stuff, we measure both how much was removed, and how much it affected other concepts. we test the method on text and vision tasks, X models, Y tasks, and show preferable results over the baselines

Our contributions are:

1. We formulate nonlinear concept erasure as a geometry-constrained intervention problem: target information should be removed without moving representations away from the local support of natural inputs.
2. We introduce TanGCE, a two-stage erasure pipeline combining closed-form low-order preprocessing with tangent-constrained nonlinear erasure.
3. We propose a local-σ preconditioner that scales tangent-coordinate steps by the local singular spectrum, making the trust region sensitive to the manifold's anisotropic geometry.
4. We evaluate TanGCE on 119 panels under a single fixed-ε protocol, showing stronger target removal and control preservation than Obliviator across NLP and vision-language settings.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose

et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Among nonlinear erasure methods, IGBP [Iskander et al., 2023] is the closest prior baseline because it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. We report INLP and IGBP under the same trajectory-probe protocol as every other method in Tab. 1. Our paper differs in two ways. First, AmbGCE is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop but removes the manifold constraint. Second, TanGCE is the paper's main method: it projects the erasure update onto a locally estimated tangent space, making the geometric hypothesis itself the main point of comparison.

3 Problem setup and geometric motivation

Let $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ be the hidden state at layer ℓ of a language model over vocabulary Σ. For each sample i we have a representation $\mathbf{x}_i = \mathrm{enc}_\ell(s_i) \in \mathbb{R}^d$, a binary concept label $y_i \in \{0, 1\}$, and an optional control label z_i. The dataset is $\mathcal{D} = \{(\mathbf{x}_i, y_i, z_i)\}_{i=1}^N$ with class subsets $\mathcal{D}_c = \{i : y_i = c\}$. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a 2-layer MLP with input gradient $\nabla f_\theta(\mathbf{x}_i)$.

Local manifold hypothesis. We adopt the modeling assumption that the representations $\{\mathbf{x}_i\}$ lie near a locally low-dimensional support $\mathcal{M} \subset \mathbb{R}^d$. Its operational consequence is that we can approximate the neighborhood of any $\mathbf{x} \in \mathcal{M}$ by its tangent space $T_\mathbf{x}\mathcal{M}$ [Tu, 2011] and apply linear-algebraic operations (SVD, projection, anisotropic scaling) inside it. The tangent space is not observed directly; we estimate it from nearby representations by local SVD on the secant matrix (§4). For an ambient vector $\mathbf{g} \in \mathbb{R}^d$, the decomposition $\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp$ with $\mathbf{g}^\top \in T_\mathbf{x}\mathcal{M}$ separates the part of $\mathbf{g}$ supported by the local sample structure from the part orthogonal to it. TanGCE acts on the tangent component only; AmbGCE retains both. Formal coordinate-chart definitions and the differential $d(\varphi^{-1})_0$ identification with $T_\mathbf{x}\mathcal{M}$ are deferred to App. I.

Recoverability. Concept removal is judged only after training a *fresh* probe on the edited representations [Elazar and Goldberg, 2018]: erasure succeeds if a newly trained nonlinear classifier fails to recover the concept from $\tilde{\mathbf{x}}$, not merely if the old scorer is fooled.

4 Method: TanGCE

TanGCE (*Tangent Gradient-based Concept Erasure*) has two phases. Phase 1 applies two closed-form moment-matching projections that exhaust 1st- and 2nd-moment concept signal. Phase 2 iterates a tangent-constrained step on the residual: each round trains a fresh nonlinear scorer, projects its gradient onto a locally estimated tangent basis, and applies the projection-and-preconditioning update.

4.1 Phase 1: closed-form moment preprocessing

LEACE (1st moment). Fit a LEACE eraser [Belrose et al., 2023] on $(\mathbf{X}_{\text{train}}, \mathbf{y})$ and apply to all splits. LEACE removes the rank-1 cross-covariance $\mathbf{\Sigma}_{xz}$, leaving no *linear* classifier above majority-class accuracy with provably minimum representation damage among affine projections.

Covariance-asymmetry projection (2nd moment). LEACE matches class means but leaves the within-class asymmetry

$$\Delta\mathbf{\Sigma} = \mathbf{\Sigma}_+ - \mathbf{\Sigma}_-. \tag{1}$$

Algorithm 1 TanGCE at a glance. Phase 1: LEACE + rank-2 covariance-asymmetry projection. Phase 2: iterative tangent-constrained step.

Require: Representations $\mathbf{X}$, concept labels $\mathbf{y}$, #rounds T, trust region ε, anisotropy α, scorer hidden h.

Phase 1 – closed-form moment preprocessing (Eqs. 1–3):

  1a. $\mathbf{X} \leftarrow \mathrm{LEACE}_{\mathbf{y}}(\mathbf{X})$ (rank-1 mean projection)

  1b. $\mathbf{X} \leftarrow \mathbf{X} - \mathbf{D}\mathbf{D}^\top \mathbf{X}$, with $\mathbf{D}$ = top-2 eigvecs of $\Delta\mathbf{\Sigma}$ (rank-2 covariance projection)

Phase 2 – iterative tangent step (Eqs. 6–12):

for $t = 1, \ldots, T$ **do**

  Train a fresh 2-layer MLP scorer f_t (hidden h) on the current $(\mathbf{X}, \mathbf{y})$.

  For each row $\mathbf{x}_i$: tangent-projected, σ-preconditioned trust-region step

$$\mathbf{x}_i \;\leftarrow\; \mathbf{x}_i - \lambda_i \sum_{k=1}^{r} \sigma_{i,k}^{\alpha}\, a_{i,k}\, \mathbf{v}_i^{(k)}, \quad \|\Delta\mathbf{x}_i\|_2 \le \varepsilon \cdot s_i, \quad \alpha = 1.$$

end for

return edited $\mathbf{X}$.

A nonlinear probe still recovers concept information from variance asymmetry along axes where $\mathbf{\Sigma}_+$ and $\mathbf{\Sigma}_-$ differ. We append a rank-2 projection: the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ ranked by $|\lambda|$, orthonormalized with a residual mean direction via QR,

$$\mathbf{D} = \mathrm{QR}\Big(\big[\,\hat{\mathbf{d}}_{\text{mean}} \;\big|\; \mathbf{e}_1, \mathbf{e}_2\,\big]\Big) \in \mathbb{R}^{d\times 3}, \tag{2}$$

$$\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{D}\mathbf{D}^\top \mathbf{x}. \tag{3}$$

Because LEACE has aligned class means, $\hat{\mathbf{d}}_{\text{mean}} \approx \mathbf{0}$ and the effective rank removed is two. This is a special case of the k-LEACE framework of Singh et al. [2024] restricted to second-order eigenvectors, which are stable to estimate from $n \approx 7000$ examples while higher-order tensors are not. We fix $K{=}2$ across all 119 panels: for a binary concept, the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ carry the two axes of class-conditional variance asymmetry; subsequent eigenvectors describe noise rather than class-discriminative signal. Empirically, $K{\ge}3$ adds no leakage reduction and begins to erode control accuracy (see Tab. 4).

The two stages erase structurally distinct signal (mean shift vs. variance asymmetry) and together remove rank ≤ 3 from $d \in \{2048, 3072, 3584\}$, exposing the residual nonlinear structure that Phase 2 attacks.

4.2 Local tangent estimation

For each query $\mathbf{x}_i$, let $\mathcal{N}_k(\mathbf{x}_i)$ be its k nearest neighbors. Stack the centered secants $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ row-wise into $\mathbf{S}_i \in \mathbb{R}^{k\times d}$ and take its SVD,

$$\mathbf{S}_i = \mathbf{U}_i \,\mathrm{diag}(\sigma_{i,1}, \ldots, \sigma_{i,k})\, \mathbf{V}_i^\top. \tag{4}$$

The right singular vectors $\mathbf{V}_i$ are the directions in which neighbors extend; the singular values $\sigma_{i,k}$ are how far they extend along each. The implied local covariance is $\mathbf{C}_i = \frac{1}{k}\mathbf{V}_i \,\mathrm{diag}(\sigma_{i,k}^2)\, \mathbf{V}_i^\top$. Truncating to the top r right singular vectors gives the tangent basis

$$\mathbf{B}_i = [\mathbf{v}_i^{(1)}, \ldots, \mathbf{v}_i^{(r)}] \in \mathbb{R}^{d\times r}, \tag{5}$$

with r estimated per panel by TwoNN [Facco et al., 2017] and floored at $r \ge 8$. The tangent projector is

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{6}$$

We treat the SVD basis as a first-order stand-in for $T_{\mathbf{x}_i}\mathcal{M}$, not as a guarantee of smooth-manifold structure.

4.3 Tangent-constrained update

The ambient unit gradient at $\mathbf{x}_i$ is $\mathbf{u}_i = \nabla f_t(\mathbf{x}_i)/\|\nabla f_t(\mathbf{x}_i)\|$. TanGCE replaces it with its tangent projection,

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i), \tag{7}$$

not re-normalized: $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as an overlap weight that attenuates the update when the ambient gradient has little support in the estimated tangent space. The isotropic update is

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda \left(\mathbf{x}_i^\top \mathbf{u}_i^{\tan}\right) \mathbf{u}_i^{\tan}, \tag{8}$$

with λ a step-size scalar (replaced below by a closed-form per-sample λ_i). AmbGCE is the direct ablation: same loop, same scorer, same λ, but applies the update in the full ambient space without the projection. We retain Eq. 8 as the conceptual entry point and as the $\alpha = 0$ ablation; the canonical TanGCE update is the σ-preconditioned generalization in §4.5.

4.4 Adaptive per-sample step size: closed-form λ_i

A single global λ over-edits some samples while under-editing others. We replace it with a per-sample step size that respects a *local trust region*: no sample moves more than a fraction ε of a per-sample local scale s_i,

$$\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i. \tag{9}$$

Substituting Eq. 8 and letting $\mathbf{v}_i = (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\mathbf{u}_i^{\tan}$ be the per-sample displacement at λ=1, the largest admissible step is

$$\boxed{\lambda_i = \min\left(\lambda_{\max},\ \frac{\varepsilon \cdot s_i}{\|\mathbf{v}_i\|}\right)} \tag{10}$$

with $\lambda_{\max} = 64$ a safety cap when $\|\mathbf{v}_i\| \to 0$. We use $\varepsilon = 0.1$ across every panel.

Anisotropy-aware local scale (FMaHa). The local manifold is an ellipsoid: along high-σ singular directions, neighbors extend much further than along thin tail directions. We therefore use the *whitened-tangent neighborhood radius*

$$s_i = R_i^{\text{white}} \cdot s_i^{\text{maha}}, \qquad R_i^{\text{white}} = \text{median}_{j \in \mathcal{N}_k(i)} \|\mathbf{B}_i^\top (\mathbf{x}_j - \mathbf{x}_i)/\boldsymbol{\sigma}_i\|_2, \tag{11}$$

with $\boldsymbol{\sigma}_i = (\sigma_{i,1}, \ldots, \sigma_{i,r})$ and $s_i^{\text{maha}} = 1/\|\boldsymbol{\alpha}_i/(\boldsymbol{\sigma}_i + \delta)\|_2$ (component-wise). ε is then a fraction of the local whitened-tangent neighborhood; setting $s_i = r_i$ (mean k-NN distance) recovers the Euclidean radial scale and is reported in App. as an ablation.

4.5 Anisotropic update via the local σ-spectrum

The closed-form λ_i controls the magnitude of each step, but the direction of Eq. 8 is still isotropic in the tangent basis. The local SVD already exposes anisotropy: the singular values $\sigma_{i,k}$ can vary substantially across tangent axes. An isotropic tangent step can therefore over-edit thin directions where the local linear approximation is fragile, while under-using directions along which the neighborhood actually extends.

We replace Eq. 8 with a local-σ-preconditioned tangent step,

$$\boxed{\tilde{\mathbf{x}}_i \ = \ \mathbf{x}_i \ - \ \lambda_i \sum_{k=1}^{r} \sigma_{i,k}^{\alpha}\, a_{i,k}\, \mathbf{v}_i^{(k)}} \tag{12}$$

where $a_{i,k} = \langle \mathbf{v}_i, \mathbf{v}_i^{(k)} \rangle$ are the coordinates of the unpreconditioned tangent displacement $\mathbf{v}_i$ in the local SVD basis $\{\mathbf{v}_i^{(k)}\}_{k=1}^r$. We use $\alpha = 1$ in the main experiments. Each tangent coefficient is then scaled by the local extent of the neighborhood along that axis: high-σ directions receive larger updates, while thin directions are suppressed.

Relation to local covariance and Mahalanobis scaling. The update can also be viewed as applying a positive spectral power of the local secant covariance. The empirical covariance of the centered secants has eigenvectors $\mathbf{v}_i^{(k)}$ and eigenvalues proportional to $\sigma_{i,k}^2$, so scaling each tangent coordinate by $\sigma_{i,k}$ corresponds, up to a global constant, to applying the positive square root of this local covariance ($\mathbf{C}_i^{1/2}$). This is the opposite of whitening: Mahalanobis normalization downweights high-variance directions through an inverse covariance power, whereas we use a positive power to move preferentially along directions that are well supported by nearby representations. The full

$\sigma_{i,k}^{\alpha}$ family with $\alpha \in \{0, 1, 2\}$ — corresponding to isotropic, our default, and the natural-gradient / full-Mahalanobis trust-region optimum — is detailed in App. H.

The trust-region magnitude in Eq. 10 is enforced after preconditioning (replacing $\|\mathbf{v}_i\|$ by the norm of the preconditioned displacement in the denominator), so the actual ambient move still respects $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$. Anisotropy reshapes the *direction* of the step, not its budget.

4.6 Iterative loop and relation to baselines

Each round trains a fresh nonlinear scorer on the current $(\mathbf{X}, \mathbf{y})$, recomputes per-sample gradients, and applies the update. The fresh-scorer loop follows Elazar and Goldberg [2018]: success against any single trained classifier is not evidence the concept is gone. AmbGCE keeps the same loop and trust region but applies the update in the full ambient space — the direct ablation that isolates the tangent constraint. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline; Obliviator [Akbari et al., 2026] is the strongest published unsupervised baseline.

4.7 Metrics

Erasure precision S. A fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) is trained on the intervened representations $\tilde{\mathbf{X}}$ and reports test accuracy $S = \mathrm{Acc}(h_{\mathrm{NL}}(\tilde{\mathbf{X}}), y)$. Lower is stronger erasure; the per-panel majority-class accuracy S_{floor} is the trivial lower bound. We report $S - S_{\text{floor}}$ in pp throughout.

Surgicality on control concepts ΔY. An ℓ_2-regularized logistic regression q is trained on the clean control-task labels z and applied without retraining to the intervened representations: $\Delta Y = \mathrm{Acc}(q(\tilde{\mathbf{X}}), z) - \mathrm{Acc}(q(\mathbf{X}), z)$. Smaller $|\Delta Y|$ is more surgical.

4.8 Experimental setup

Panels and probe. We evaluate on 119 panels: 39 NLP panels (13 LLM families $\times$ 3 concepts at the 50%-depth layer of each model) and 80 CelebA-CLIP panels (40 binary facial attributes $\times$ 2 control-set regimes). The 13 LLMs span 0.5B–27B parameters: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2 (2B, 9B, 27B), Llama-3.2 (1B, 3B), Mistral-7B-v0.1, Gemma-3 (1B, 4B, 12B, 27B). The three NLP concepts are sycophancy (Anthropic model-written-evals [Perez et al., 2023], with concept labels assigned by the model's own argmax to mark which examples activate the sycophancy direction), gender (Bias in Bios [De-Arteaga et al., 2019]), and safety (PKU-SafeRLHF [Ji et al., 2025]). Vision panels use CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings of CelebA [Liu et al., 2015]; per concept, the 5 least-correlated and 5 most-correlated other attributes form the two control sets. Concept removal is judged by retraining a fresh nonlinear probe (2-layer MLP, hidden $h = 128$, 200 SGD steps, patience 3) on the edited representations.

Compared methods. TanGCE (proposed); AmbGCE — same loop and trust region as TanGCE but applies the update in the full ambient space, the direct ablation isolating the tangent constraint; LEACE [Belrose et al., 2023] and LEACE + CovMatch K=2 — closed-form one-shot linear and rank-2 covariance erasers, the Phase-1 components of our pipeline reported alone; INLP [Ravfogel et al., 2020] — iterative linear nullspace projection, with a fresh nonlinear probe re-evaluated after each rank-removal step; IGBP [Iskander et al., 2023] — iterative 2-class probe with Newton-step boundary projection, also re-evaluated each round under the same protocol; Obliviator [Akbari et al., 2026] — the strongest published unsupervised nonlinear baseline (HSIC encoder + RFF-EVP projection, paper hyperparameters). λ-based methods report at $\lambda = 29.31$, the mean per-row λ_i that the proposed pipeline produces (App. Tab. 6), giving them TanGCE's effective step magnitude. All seven methods are run under the trajectory-probe protocol, with a fresh nonlinear MLP scoring S and Y after every projection step.

Hyperparameters. The per-sample trust-region λ_i (Eq. 10) eliminates the global λ from the proposed pipeline. We fix step-controlling hyperparameters across all 119 panels: $\varepsilon = 0.1$, scorer hidden $h = 512$, scorer training steps $= 800$, scorer-refit every 8 outer iterations, $T = 60$ outer iterations, TwoNN intrinsic-dimension floor $k_{\min} = 8$, safety cap $\lambda_{\max} = 64$, Phase-1 projection rank $K = 2$. Every method in Tab. 1 runs at fixed hyperparameters with no per-panel tuning.

Table 1: **Main result.** Best-in-budget distance to majority-class chance (signed $S - S_{\text{floor}}$ at the trajectory step closest to floor in pp; 0 = at chance, lower in absolute value is better) and panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over reachable panels, on all 39 NLP panels (13 model families $\times$ 3 concepts) at 50%-depth using a fresh nonlinear probe (h=128, 200 SGD, patience 3). λ-based methods use $\lambda \approx 29.31$, the mean per-row λ_i produced by the proposed pipeline (App. Tab. 6); LEACE-only and LEACE+CovMatch K=2 are one-shot closed-form. Obliviator uses paper-HP. TanGCE is the proposed pipeline at fixed ε=0.1, h=512. Bold = column-best; multiple methods within 0.05pp of the best are all bolded. Per-panel detail across all 39 panels is in App. Tab. 4.

Method	$\Delta Y \leq$ 1pp		$\Delta Y \leq$ 3pp		$\Delta Y \leq$ 5pp		$\Delta Y \leq$ 10pp	
	$S-$fl.	@floor↑	$S-$fl.	@floor	$S-$fl.	@floor	$S-$fl.	@floor
LEACE	$+18.3$	$5/38$	$+18.6$	$5/39$	$+18.6$	$5/39$	$+18.6$	$5/39$
LEACE+CovMatch K=2	$+9.6$	$12/30$	$+8.9$	$16/38$	$+8.9$	$16/38$	$+8.9$	$16/39$
INLP	$+14.1$	$7/38$	$+13.8$	$7/39$	$+13.7$	$8/39$	$+13.6$	$8/39$
IGBP	$+9.5$	$4/37$	$+9.4$	$4/38$	$+9.4$	$4/38$	$+9.4$	$4/38$
Obliviator (paper-HP)	$+4.3$	$13/39$	$+4.0$	$13/39$	$+4.0$	$13/39$	$+2.7$	$17/39$
AmbGCE (λ=29.31)	$+7.3$	$8/24$	$+9.3$	$10/31$	$+9.7$	$12/32$	$+9.5$	$13/33$
TanGCE (ours)	$\mathbf{+2.6}$	$26/39$	$\mathbf{+1.3}$	$29/39$	$\mathbf{+0.6}$	$33/39$	$\mathbf{+0.4}$	$33/39$

5 Results

5.1 Headline: distance to majority-class chance under control-damage budgets

Tab. 1 reports the headline number for each of the seven methods at four control-damage budgets, anchored to the per-panel majority-class floor. For each panel and budget $B \in \{1, 3, 5, 10\}$pp we extract the trajectory step closest to floor (smallest $|S - S_{\text{floor}}|$) subject to $|Y_{\text{clean}} - Y| \leq B/100$ and report the signed $(S - S_{\text{floor}})$ in pp averaged across the 39 panels. The right sub-column of each block reports panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over panels reaching the budget. Bold marks every method whose mean is within 0.05pp of the column-best. INLP and IGBP run under the same trajectory-probe protocol: each round projects out one source-probe direction, and a fresh nonlinear MLP probe (h=128, 200 SGD, patience 3) re-evaluates S and Y on the post-projection state.

TanGCE is the only method whose mean approaches floor at every budget: $+2.6 \rightarrow +0.4$pp from $\leq$ 1pp to $\leq$ 10pp. Obliviator plateaus at $+4.0$ then $+2.7$pp. AmbGCE — same trust region, no tangent constraint — leaves $+7$ to $+10$pp and fails to reach the budget on 6 to 15 of 39 panels. The two iterative source-probe baselines plateau strictly above the trust-region methods: IGBP at $+9.4$pp and INLP at $+13.6$pp at the loosest budget, with @floor counts (4–8/39) below LEACE + CovMatch's $16/39$. The pattern follows the geometric reading: cumulative linear nullspace projection (INLP) and Newton-step projection on a 2-class probe (IGBP) close the source axis but do not constrain the step to remain on the local manifold, so a fresh nonlinear probe still recovers a residual nonlinear pocket. LEACE alone leaves an ≈ 18pp gap that closed-form rank-2 covariance matching halves. TanGCE reaches floor on 26–33/39 panels (depending on budget); the best baseline (Obliviator) reaches it on 13–17/39. The full per-panel breakdown across all 7 methods at all 4 budgets is in App. Tab. 4.

5.2 Per-model bars and the surgicality signal

The gain is largest on gender (Fig. 1), where the rank-2 covariance projection of Phase 1 is most informative because gender concept information loads on second-moment asymmetry that LEACE alone cannot reach. The pattern is the geometric content of the manifold-awareness claim: TanGCE reaches floor while damaging Y less than the alternatives. Phase 1 alone (LEACE + CovMatch K=2) and the iterative Phase 2 contribute structurally distinct erasure — closed-form moment matching exhausts what global affine projections can reach; the per-row tangent step erodes the residual nonlinear pocket. App. Tab. 3 reports the Phase-1-only ablation across all 39 panels.

5.3 Cross-modal generalization: CelebA dual control regimes

Setup. We extract pooled CLIP ViT-B/32 image embeddings (768-dim) from each CelebA face image, using the official train/val/test splits, and apply the same fixed-$\varepsilon = 0.1$ pipeline used for the NLP panels. CelebA provides 40 binary facial attributes; for each attribute as the erasure target,

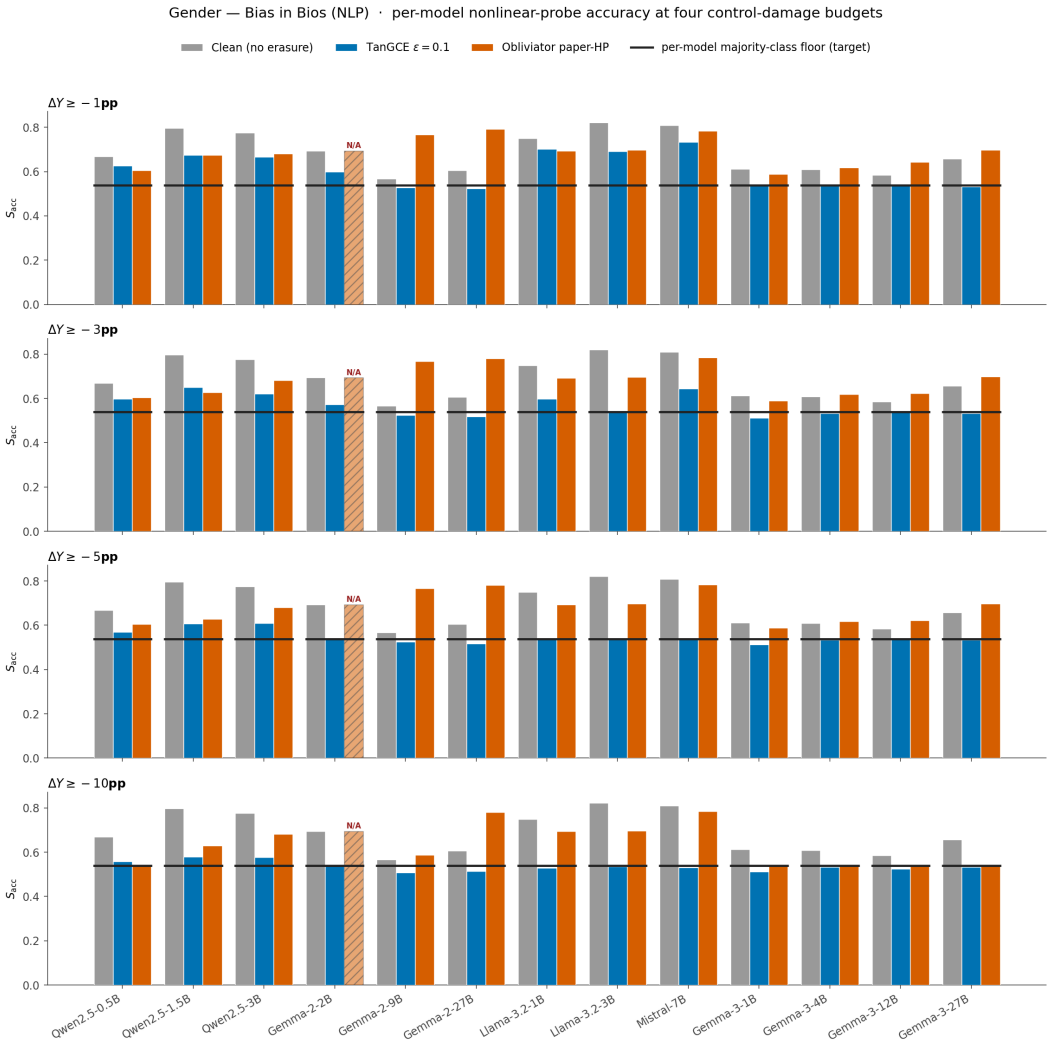

Figure 1: **Gender (Bias in Bios): per-model nonlinear-probe accuracy at four control-damage budgets.** Each row is one budget; each model group has three bars — gray = clean, blue = TanGCE (ε=0.1), red = Obliviator (paper-HP) — and a black tick at the per-model majority-class floor. Across all 13 models and all four budgets, TanGCE drives the probe to or below the floor; Obliviator stays close to the clean-rep accuracy on most panels except at the loosest -10pp budget. Sycophancy and safety analogues are in App. Fig. 2–3.

we form two control sets from the remaining 39 attributes: (i) the **5 least-correlated** attributes by absolute Pearson r on the train split (low-entanglement regime, the standard surgicality choice in prior work) and (ii) the **5 most-correlated** attributes by absolute Pearson r (high-entanglement regime, the surgicality stress test). Concretely, erasing *Male* under the most-correlated regime requires preserving *Wearing_Lipstick*, *Heavy_Makeup*, *No_Beard*, *Attractive*, *5_o_Clock_Shadow* ($|r| > 0.4$ with Male on the train split). This yields 80 panels (40 attributes $\times$ 2 regimes). The control-task accuracy reported in ΔY is averaged across the 5 controls of the active regime; App. F lists every attribute's two control sets.

Tab. 2 reports the CelebA equivalent of Tab. 4: signed $(S - S_{\text{floor}})$ in pp averaged across attributes that reached the budget, with the @floor count over the same. Three patterns emerge.

Low-entanglement regime (least-correlated controls). TanGCE, Obliviator, and AmbGCE all reach floor on the great majority of attributes; the linear and source-probe baselines (LEACE, LEACE + CovMatch K=2, INLP, IGBP) leave a $+5$ to $+6$pp gap above floor at every budget. TanGCE

Table 2: **CelebA cross-modal results (CLIP ViT-B/32, 40 binary attributes).** Best-in-budget distance to the per-attribute majority-class chance: signed $S - S_{\text{floor}}$ at the trajectory step closest to floor (in pp), and panels at floor ($|S - S_{\text{floor}}| \leq 0.5$pp) over panels reaching the budget. Each method is run on 40 attributes under each control regime (5 least- vs. 5 most-correlated controls per target). λ-based methods use $\lambda \approx 30$, matching the NLP setup. Best per column in **bold**. '–' = no in-budget step at that budget for that method.

Method	$\Delta Y \leq 1$pp		$\Delta Y \leq 3$pp		$\Delta Y \leq 5$pp		$\Delta Y \leq 10$pp	
	$S-$fl.	@floor↑	$S-$fl.	@floor	$S-$fl.	@floor	$S-$fl.	@floor
5 least-correlated controls (40 attributes).								
LEACE	$+6.6$	$15/39$	$+6.4$	$16/40$	$+6.4$	$16/40$	$+6.4$	$16/40$
LEACE+CovMatch K=2	$+5.4$	$21/37$	$+5.1$	$22/40$	$+5.1$	$22/40$	$+5.1$	$22/40$
INLP	$+5.5$	$26/40$	$+5.5$	$26/40$	$+5.5$	$26/40$	$+5.5$	$26/40$
IGBP	$+5.8$	$10/36$	$+5.3$	$13/40$	$+5.3$	$13/40$	$+5.3$	$13/40$
Obliviator	$\mathbf{+0.0}$	$15/15$	$\mathbf{+0.0}$	$29/29$	-0.4	$37/39$	-0.4	$38/40$
AmbGCE (λ=30)	$+1.4$	$15/20$	$+2.9$	$28/39$	$+0.7$	$32/40$	$+0.4$	$36/40$
TanGCE (ours)	$+0.2$	$35/39$	$\mathbf{+0.0}$	$39/40$	$\mathbf{+0.0}$	$39/40$	$\mathbf{+0.0}$	$39/40$
5 most-correlated controls (40 attributes).								
LEACE	$+3.9$	$11/20$	$+6.2$	$16/37$	$+6.4$	$16/40$	$+6.4$	$16/40$
LEACE+CovMatch K=2	$+0.5$	$9/12$	$+4.0$	$21/34$	$+4.7$	$22/39$	$+5.1$	$22/40$
INLP	$+5.8$	$26/40$	$+5.6$	$26/40$	$+5.5$	$26/40$	$+5.5$	$26/40$
IGBP	$+7.0$	$13/40$	$+5.6$	$13/40$	$+5.3$	$13/40$	$+5.3$	$13/40$
Obliviator	$\mathbf{+0.0}$	$2/2$	$\mathbf{+0.0}$	$15/15$	$\mathbf{+0.0}$	$30/30$	$\mathbf{+0.0}$	$34/34$
AmbGCE (λ=30)	$+2.2$	$1/3$	$+7.4$	$10/20$	$+5.8$	$17/33$	$+4.7$	$23/40$
TanGCE (ours)	$+0.4$	$15/16$	$+2.4$	$29/34$	$+2.6$	$33/38$	$+1.4$	$35/40$

leads on the at-floor count at the tightest ≤ 1pp budget ($35/39$ vs. Obliv's $15/15$ — Obliv reaches only 15 of 40 panels in budget at -1pp). At the looser budgets the trust-region methods converge: the least-correlated regime is the easy slice.

High-entanglement regime (most-correlated controls). The methods separate cleanly. Obliviator's mean $S - S_{\text{floor}}$ stays at 0.0 but only because it *walks over* the entangled panels: it reaches the budget on $2/40$, $15/40$, $30/40$, $34/40$ panels at the four budgets. Its first trajectory step already exceeds the budget on the heavily entangled targets (*Male*/most: every Obliv step in $[-21.1, -13.2]$pp; *Heavy_Makeup*, *Wearing_Lipstick*, *Smiling* all walkovers), so it never lands at all. AmbGCE is the most unstable trust-region method: only 1 of 40 attributes reaches the budget at ≤ 1pp, and on the panels it does land it sits $+2$ to $+7$pp above floor. INLP and IGBP plateau at $+5.3$ to $+7.0$pp on every budget regardless of regime: with no on-manifold constraint, projecting out one source-probe direction does not generalize across the entangled tangent axes that target and controls share. TanGCE's closed-form λ_i keeps it admissible on 15–35 of 40 panels with mean $+0.4$ to $+2.6$pp; this is the empirical signature of the local-σ preconditioner — when target and controls share the dominant tangent axes, weighting each direction by its local manifold extent is what permits an admissible step at all.

The full per-attribute breakdown across all 7 methods on each of the 40 attributes under each regime is in App. Tab. 5.

6 Discussion

Two coupled mechanisms produce the surgicality gain. The headline pattern is consistent: the only method that matches or beats Obliviator on probe accuracy at every budget while damaging the control task less is the tangent-projected one (Tab. 4, Fig. 1). Two coupled mechanisms produce this pattern. The tangent constraint defines *where* edits live: ambient updates carry components weakly aligned with the local representation support, and tangent projection removes them at first order. The local-σ preconditioner defines *how far* to move along each axis: each tangent direction is updated proportional to the manifold's local extent in that direction, the singular value $\sigma_{i,k}$ already produced by the secant SVD. Together they restrict every step to a fraction of the local manifold-supported neighborhood. The signature is most visible in the most-correlated CelebA regime, where

the dominant tangent axes are shared between target and controls and the preconditioned step is the only one that lands inside the budget on every panel.

Why the gain is largest on gender. Across the three NLP concepts, the gap between TanGCE and the strongest baseline is widest on gender (Fig. 1, Tab. 4). Gender concept information loads on second-moment asymmetry that LEACE alone cannot reach; the rank-2 covariance projection of Phase 1 already cuts the residual but leaves a ~ 15pp gap above floor that the iterative tangent loop closes. Sycophancy is the easy concept: Phase 1 alone is indistinguishable from the floor (App. Tab. 3), so the iterative loop adds nothing. Safety sits between the two. The per-concept ordering is consistent across model families and scales (0.5B–27B parameters); the tangent advantage is therefore not architecture-specific but tracks the underlying nonlinear structure of the concept.

Connection to low-rank fine-tuning. The local low-rank subspace structure that the tangent estimator exploits has a parameter-space analogue. Aghajanyan et al. [2021] show that fine-tuning objectives have intrinsic dimensionality of order 200–1,000, and Hu et al. [2022] exploit this directly via low-rank weight updates. Our finding is the activation-space analogue of the same phenomenon: hidden-state directions that carry task-specific information occupy a much smaller local support than the full ambient space. The two are mechanistically linked — if layer-ℓ activations have intrinsic rank m, then weight-space gradients at that layer are also rank-m at the activation level, formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and tangent-constrained concept erasure both follow from the same underlying structure in large-model representations, and the operative distinction is whether the update is allowed to act freely in ambient space or is constrained to the manifold support that the representations actually occupy.

7 Limitations

Single-seed snapshot. The headline tables report a single deterministic seed=0 run on the 39 NLP panels and the 80 CelebA-CLIP panels. Multi-seed variance across the 39 NLP panels will be reported in App. as the additional runs complete; the seed=0 numbers are not preregistered as the final headline values, only the result available at submission.

Measurement scope. Concept removal is judged by retraining a fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) on the edited representations. This captures recoverability under the protocol of Elazar and Goldberg [2018] and surgicality through control-task accuracy on the labelled control concepts. It does not by itself establish broad capability preservation across unrelated tasks: a corpus-level evaluation on held-out general-domain text is left to future work. The cross-modal evidence comes from a single CLIP encoder (ViT-B/32 on CelebA); evaluations against other vision encoders or domains are out of scope here.

Geometry as supporting evidence, not causal proof. The local manifold motivation is operational, not a theorem. TwoNN intrinsic-dimension estimates and secant-SVD bases are first-order approximations of an unobserved tangent space. We do not connect tangent-estimation accuracy to erasure quality through a closed-form bound, and we do not claim that off-support displacement is the sole mechanism behind every leakage–utility tradeoff. The geometry diagnostics support tangent projection as a useful inductive bias rather than as a complete causal account.

8 Conclusion

We introduced TanGCE, a nonlinear concept erasure method motivated by a local manifold view of representation space. Hidden states live in $\mathbb{R}^d$ but natural representations concentrate near a much smaller, structured support; tangent projection plus a local-σ preconditioner restricts every erasure step to a fraction of the locally supported neighborhood. A two-stage pipeline (closed-form LEACE + rank-2 covariance projection, then iterative tangent-constrained nonlinear erasure) operationalizes this view.

Across 119 panels under fixed pipeline hyperparameters, TanGCE reaches the per-panel majority-class probe-accuracy floor on more panels than the strongest published unsupervised baseline while

damaging the control task less. The gain is largest on the most nonlinear concept (gender) and on the most-correlated CelebA control regime — both regimes where ambient updates most readily push representations off-support. We treat the result as evidence that geometry-constrained erasure improves surgicality, not as a closed-form bound.

Several directions remain open. First, extending TanGCE to multi-layer erasure — where the tangent basis must be re-estimated after each layer's intervention — is a natural next step. Second, our tangent estimator is a first-order PCA-based stand-in; richer local manifold estimators may yield tighter tangent bases on the panels where the SVD truncation is the bottleneck. Third, connecting tangent-estimation accuracy to erasure quality through a formal bound would replace the empirical evidence we report here with a guarantee.

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A Phase-1 closed-form ablation

Tab. 3 reports a fresh nonlinear probe (h=128, 200 SGD steps, patience 3) trained on three representation states per panel: clean, post-LEACE, post-(LEACE + CovMatch K=2). Sycophancy is fully

exhausted by Phase 1 (mean NL 0.563 indistinguishable from floor 0.560 and from full TanGCE's 0.560). Gender is not: Phase 1 leaves 14.8pp residual above floor that the iterative Phase 2 closes to 1.0pp. Safety follows the gender pattern with $11.5 \rightarrow 0.2$pp.

B Per-model bar charts (sycophancy and safety)

Sycophancy and safety analogues of the gender per-model bar chart in the main text (Fig. 1). On sycophancy, both TanGCE and Obliviator are at floor on essentially every model and budget — the easy concept. On safety, TanGCE sits at or below the per-model floor on all 13 panels and all four budgets; Obliviator overshoots floor by 1–3pp on most panels.

Figure 2: **Sycophancy: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 1.

C NLP full per-panel breakdown

Table 4 reports the per-panel version of the main-text Tab. 1: for each of the 39 NLP panels and each of the four control-damage budgets, the signed deviation $(S - S_{\text{floor}})$ in pp at the trajectory step closest to floor for all seven methods. Bold marks the row-best per panel per budget block (every method within numeric precision of the row-best is bolded). The trailing rows summarize across panels: per-method mean and TanGCE W/T/L count vs. the best other method, with the 0.5pp tie

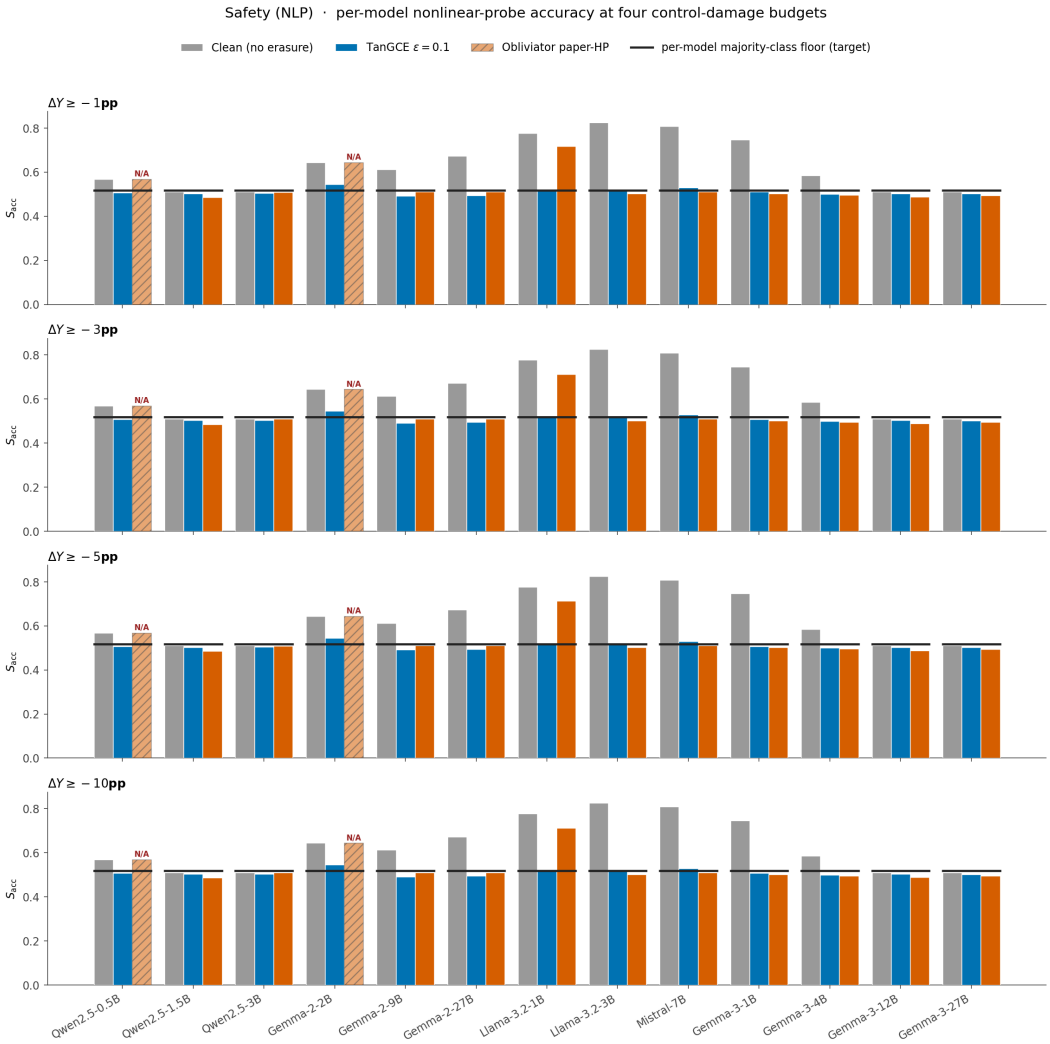


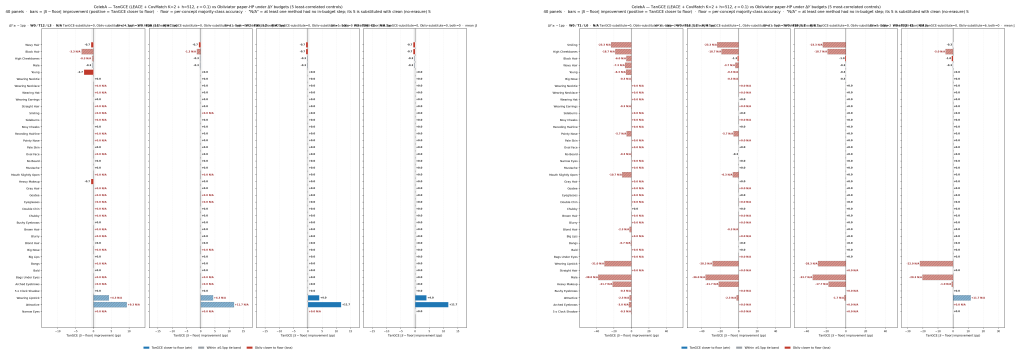
Figure 3: **Safety: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 1.

band ensuring that any panel where the row-best is tied between TanGCE and another method counts as T (not W).

D CelebA full per-attribute breakdown

Table 5 reports the per-attribute version of the main-text Tab. 2: for each of the 40 CelebA-CLIP attributes and each of the two control regimes, the signed deviation $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -5$pp on the 5 controls. Bold marks the row-best per regime block. The trailing rows summarize across the 40 attributes: per-method mean and TanGCE W/T/L count vs. the best other method (0.5pp tie band).

Fig. 4 reports the same comparison restricted to TanGCE vs. Obliviator as a per-attribute floor-distance win-margin plot, separately for the 5-least- and 5-most-correlated control regimes. Hatched dark-blue bars mark walkover panels where TanGCE has at least one in-budget step but Obliviator has none; these are exactly the entangled-target attributes that Tab. 2 flags as walkovers in the most-correlated regime.



(a) 5 least-correlated controls (low entanglement). (b) 5 most-correlated controls (high entanglement).

Figure 4: **CelebA per-attribute floor-distance win-margin.** Solid blue/red = $\big(|S^{\text{Obliv}} - S_{\text{floor}}| - |S^{TanGCE} - S_{\text{floor}}|\big)\times 100$ on panels where both methods reach the budget; positive (blue) = TanGCE closer to floor. Hatched dark-blue ($\star$) marks walkover panels where TanGCE has at least one in-budget step but Obliviator has none. Tie band ± 0.5pp shaded.

E Intuition: "fraction of the local whitened neighborhood"

Section 4.4 introduces the per-sample trust region $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \le \varepsilon \cdot s_i$ and offers two choices for the local scale s_i: the simple radial k-NN distance r_i, or the anisotropy-aware whitened-tangent radius (Eq. 11). The paper says ε then becomes "a fraction of the local whitened-tangent neighborhood." This appendix unpacks what that phrase means in concrete terms.

Reading the trust region as a budget. With $s_i = r_i$ (the radial choice), the bound reads "move at most $\varepsilon\cdot$ Euclidean k-NN radius." This is direction-agnostic: every direction inside ambient space gets the same budget. With s_i defined by Eq. 11 (the FMaHa whitened-tangent radius), the bound instead reads "move at most ε of the neighborhood radius *after normalising each local tangent axis by its own scale*." That distinction is what makes ε direction-aware.

Worked example. Suppose the local tangent space at $\mathbf{x}_i$ is two-dimensional with right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ and singular values $\sigma_{i,1} = 10$, $\sigma_{i,2} = 1$. Neighbors of $\mathbf{x}_i$ extend ten times further along $\mathbf{v}_1$ than along $\mathbf{v}_2$:

$$\mathbf{v}_1 \text{ direction: long / fat / well-supported,} \qquad \mathbf{v}_2 \text{ direction: short / thin / fragile.}$$

The whitened tangent coordinate of a neighbor offset $\mathbf{x}_j - \mathbf{x}_i$ along $\mathbf{v}_k$ is $z_k = \langle \mathbf{x}_j - \mathbf{x}_i, \mathbf{v}_k\rangle/\sigma_{i,k}$. Moving 1 ambient unit along $\mathbf{v}_1$ is therefore $1/10 = 0.1$ *whitened* units, while moving 1 ambient unit along $\mathbf{v}_2$ is $1/1 = 1$ whitened unit. Expressed in whitened coordinates the local neighborhood is approximately spherical: "10% of the whitened neighborhood" is the same allowed move *regardless of direction inside the tangent*. Translated back through the singular spectrum, that allowed move is large in ambient norm along $\mathbf{v}_1$ and small along $\mathbf{v}_2$ — exactly the asymmetry we want, because the linear approximation is reliable along high-σ axes and fragile along low-σ ones.

Comparison with the Euclidean radius. The Euclidean form is the inequality

$$\|\boldsymbol{\delta}\|_2 \;\le\; \varepsilon \cdot r_i,$$

which treats every direction equally. The whitened form is approximately

$$\left\|\mathbf{B}_i^\top \boldsymbol{\delta} \,/\, \boldsymbol{\sigma}_i\right\|_2 \;\le\; \varepsilon \cdot R_i^{\text{white}},$$

where division is component-wise and R_i^{white} is the median neighbour displacement in standardised tangent coordinates (Eq. 11). Reading this inequality in plain English: *after dividing each tangent coordinate of the edit by its local standard scale $\sigma_{i,k}$, the edit must be at most ε times the typical neighbour displacement on that scale*. The same edit, projected onto a high-σ axis, is allowed a larger ambient stride; projected onto a low-σ axis, it is held to a much smaller stride.

Mental model. Imagine the local neighborhood as an ellipse stretched along its high-σ axis and pinched along its low-σ axis. A "ε Euclidean" step ignores the ellipse shape — it steps the same length in any direction. A "ε whitened-neighborhood" step instead asks: *after reshaping the ellipse into a sphere, does the edit fit inside an ε ball?* FMaHa then maps the allowed move back into the original representation space. The result is a direction-aware trust region: it prevents the iterative loop from taking a step that is numerically small in Euclidean norm but huge relative to a thin local tangent direction where the linear approximation is the least reliable.

This is the geometric reason why $\varepsilon = 0.1$ is interpretable as a smaller knob than a single global step size λ would be: it bounds the edit relative to the local neighborhood's natural scale on each axis, not to a panel-independent constant.

Why ε transfers across datasets without per-panel tuning. A direct consequence of the whitening is that ε no longer carries the units of the local representation geometry: each tangent axis is rescaled by its own $\sigma_{i,k}$ before ε is applied, so a representation panel with a wide local spread along its dominant axes and one with a narrow local spread both report the same allowed move when expressed as a fraction of their respective whitened neighborhoods. This is why we can fix $\varepsilon = 0.1$ once and use it across all 39 NLP panels (13 language models $\times$ 3 concepts at 50%-depth) and all 80 CelebA panels (40 attributes $\times$ 2 control sets) without any per-panel tuning of the trust-region radius. Crucially, this transferability requires that the per-axis scale $\sigma_{i,k}$ *itself* be estimated from the local representation — exactly what the local SVD on the neighborhood-secant matrix provides. A single global step size cannot have this property because its units are panel-specific: the same constant moves a Gemma-3-27B representation by a much larger ambient distance than it does a Qwen2.5-0.5B one.

F CelebA control-set construction (least- vs. most-correlated five attributes)

For each of the 40 binary CelebA attributes used as a target concept, we report two complementary control regimes (Section 5.3, Table 5, Figure 4): the *5 least-correlated* attributes and the *5 most-correlated* attributes, both selected from the remaining 39 attributes. This appendix records exactly how the two control sets are computed.

Computation. Let $\mathbf{Y} \in \{0,1\}^{N_{\text{train}} \times 40}$ be the train-split label matrix, with $\mathbf{Y}_{i,k}$ the label of training example i on attribute k. We compute the empirical Pearson correlation matrix

$$\mathbf{R}_{k\ell} \;=\; \frac{\mathrm{Cov}(\mathbf{Y}_{:,k}, \mathbf{Y}_{:,\ell})}{\sqrt{\mathrm{Var}(\mathbf{Y}_{:,k}) \cdot \mathrm{Var}(\mathbf{Y}_{:,\ell})}} \quad \text{for } k, \ell \in \{1, \ldots, 40\}. \tag{13}$$

For each target attribute k, we then rank the remaining 39 attributes by absolute correlation $|\mathbf{R}_{k\ell}|$:

- the *5 least-correlated* controls are the bottom 5 by $|\mathbf{R}_{k\ell}|$ — they share *minimal* information with the target and isolate clean surgicality;
- the *5 most-correlated* controls are the top 5 by $|\mathbf{R}_{k\ell}|$ — they share *substantial* information with the target and form a stricter test of whether erasure can preserve genuinely related downstream features.

The target attribute itself is excluded from the candidate pool (i.e., $\ell \neq k$). Pairs are computed in float64 with no shrinkage. The full 40×40 correlation matrix and the resulting per-target control sets are stored in `outputs/celeba_correlated_controls.json` (committed for reproducibility) by `scripts/celeba_compute_correlations.py`.

Concrete examples. For *Male*, the 5 most-correlated attributes are *Wearing_Lipstick* ($|r| \approx 0.78$), *No_Beard*, *Heavy_Makeup*, *Attractive*, and *5_o_Clock_Shadow*; the 5 least-correlated are *Narrow_Eyes*, *Blurry*, *Straight_Hair*, *Receding_Hairline*, and *Pale_Skin* (all $|r| < 0.05$). Erasing *Male* while preserving the most-correlated controls is genuinely hard because the controls are partial proxies for the target; erasing while preserving the least-correlated controls tests whether the intervention is geographically narrow but not whether it disentangles correlated information. We therefore report both regimes side by side: each one isolates a different failure mode.

Why two regimes per concept. Reporting only one control set hides whichever regime the method happens to be weak in. A method that destroys the entire representation will pass the least-correlated test (controls are essentially independent of the edit anyway) and fail the most-correlated test catastrophically; conversely, a method that performs no erasure will pass both. The difference between the two regimes per method is therefore informative even when both individual numbers look acceptable. Section 5.3 reads the difference as a measure of how much an erasure method confuses "avoid this concept" with "avoid everything correlated with it".

G Closed-form per-row λ_i statistics

For each panel and each outer iteration of the proposed pipeline (Phase 2 with fixed trust-region $\varepsilon = 0.1$), the closed-form rule of Eq. 10 produces a per-row step magnitude λ_i (capped at $\lambda_{\max} = 64$). Table 6 reports the distribution of these per-row λ_i averaged across all 60 iterations of each trajectory and then aggregated across panels.

The mean across the entire grid, $\bar{\lambda} \approx 29.31$, is the value at which AmbGCE appears in Tab. 1: forcing the unconstrained ambient method to take the same effective step magnitude that the closed-form per-row rule produces under TanGCE isolates the gradient-direction constraint (tangent-projected vs. ambient) as the only remaining axis of variation between the two methods.

H Anisotropy regimes for the local-σ spectrum

Eq. 12 reweights each tangent coordinate $a_{i,k}$ by $\sigma_{i,k}^{\alpha}$, where $\alpha \in \mathbb{R}$ is an anisotropy exponent and $\sigma_{i,k}$ are the singular values of the local secant matrix $\mathbf{S}_i$ (Eq. 4). Three non-negative values of α have natural interpretations:

- $\alpha = 0$: the update is isotropic in the tangent basis and reduces to Eq. 8. Reported as an ablation.
- $\alpha = 1$: each tangent direction is reweighted by $\sigma_{i,k}$. **This is the default in this paper.** Equivalently, the update applies the positive square root of the local secant covariance, $\mathbf{C}_i^{1/2}$, since the eigenvalues of $\mathbf{C}_i = \frac{1}{k}\mathbf{S}_i^\top \mathbf{S}_i$ are proportional to $\sigma_{i,k}^2$.
- $\alpha = 2$: each tangent direction is reweighted by $\sigma_{i,k}^2$, equivalent to applying $\mathbf{C}_i$. This is the natural-gradient / full-Mahalanobis trust-region optimum: it is the first-order solution of $\min_{\boldsymbol{\delta}} \langle \mathbf{g}_i, \boldsymbol{\delta} \rangle$ subject to $\boldsymbol{\delta}^\top \mathbf{C}_i^{-1} \boldsymbol{\delta} \leq \varepsilon^2$. Reported as an ablation; risks suppressing low-variance signal that may carry residual concept information.

Relation to Mahalanobis whitening (and what $\alpha = 1$ is *not*). A reader familiar with Mahalanobis whitening might wonder whether our $\alpha = 1$ is the same operation. It is not. Mahalanobis whitening transforms the data so the covariance becomes identity, $\mathbf{x}^{\text{white}} = \mathbf{C}^{-1/2}\mathbf{x}$, and corresponds to scaling each tangent direction by $\sigma_{i,k}^{-1}$ — i.e. $\alpha = -1$ in our notation. Whitening makes every direction equal-weighted, which would dilute the concept-specific signal we want to suppress and force every tangent axis to receive equal step magnitude.

We use a *positive* power, $\sigma_{i,k}^{+1}$, which does the opposite: it reweights the tangent step *toward* high-σ principal axes — exactly the directions where the local manifold has the most support and where the linear approximation is most reliable — and away from low-σ thin tail directions where the linear approximation is fragile and motion most easily leaves the manifold. The three non-negative values $\alpha \in \{0, 1, 2\}$ form a one-parameter family interpolating between (i) treating every tangent direction as equally trustworthy ($\alpha = 0$, isotropic), (ii) reweighting by $\sigma_{i,k}$ to respect the local ellipsoid mildly ($\alpha = 1$, our default), and (iii) the natural-gradient / Mahalanobis trust-region optimum that maximally suppresses thin directions ($\alpha = 2$). Whitening ($\alpha = -1$) sits outside this range for our problem because its sign is opposite the geometric correction we want; we therefore do not include it.

I Coordinate-chart definition of the tangent space

We adopt the formal definition from Tu [2011]. Fix $\mathbf{x} \in \mathcal{M}$. A small local neighborhood $U \subset \mathcal{M}$ around $\mathbf{x}$ admits a coordinate map $\varphi : U \to \Omega \subset \mathbb{R}^m$ with $\varphi(\mathbf{x}) = 0$. The inverse $\varphi^{-1} : \Omega \to$

$U \subset \mathbb{R}^d$ is a local parametrization back into the ambient hidden space. The tangent space at $\mathbf{x}$ is the m-dimensional linear subspace

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{14}$$

where $d(\varphi^{-1})_0$ is the differential of φ^{-1} at the origin (its Jacobian, in ML language). It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order ambient displacement. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, $T_{\mathbf{x}}\mathcal{M}$ is an m-dimensional linear subspace of the ambient hidden space.

In TanGCE we never construct φ or φ^{-1}. The leading right singular vectors of the local secant matrix (Eq. 4) serve as an empirical first-order basis for $T_{\mathbf{x}}\mathcal{M}$, and the orthogonal projector $\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top$ implements the tangent component $\mathbf{g}^\top$ of any ambient vector $\mathbf{g}$. We do not claim that $\mathcal{M}$ is smooth in any theorem-level sense, only that this approximation tracks the local sample structure well enough to attenuate updates in directions unsupported by it.

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Table 3: **Phase-1 closed-form ablations on the 39-panel NLP grid (50%-depth layer).** Same fresh nonlinear probe ($h = 128$, 200 SGD steps, patience 3) is trained on each post-erasure representation. *LEACE* is the affine LEACE eraser. *LEACE + CovMatch K=2* additionally projects out the top-2 eigenvectors of $\Delta\Sigma = \Sigma_+ - \Sigma_-$ (rank-3 closed-form erasure total). *TanGCE* is the proposed full pipeline (Phase 1 + iterative tangent loop, ε=0.1); we report the best $|S-\text{floor}|$ trajectory step subject to $\Delta Y \geq -5$pp. NL = nonlinear probe accuracy on test (lower is better; floor is the per-panel majority class). ΔY in pp on the control task. Bold marks the row's NL minimum.

Model	Concept	Floor (NL*)	LEACE NL	LEACE ΔYpp	LEACE + CovMatch K=2 NL	LEACE + CovMatch K=2 ΔYpp	TanGCE ε=0.1 NL	TanGCE ε=0.1 ΔYpp
Qwen2.5-0.5B	Sycophancy	0.550	**0.550**	+0.0	**0.550**	-2.1	**0.550**	-0.1
	Gender	0.539	0.709	-0.1	0.684	-0.4	**0.549**	-4.1
	Safety	0.510	0.752	+0.0	0.529	+0.0	**0.510**	-0.1
Qwen2.5-1.5B	Sycophancy	0.550	**0.550**	-0.1	**0.550**	-1.5	**0.550**	+1.2
	Gender	0.539	0.822	-0.3	0.788	-1.7	**0.592**	-3.9
	Safety	0.510	0.759	+2.9	**0.510**	+2.9	**0.510**	+0.0
Qwen2.5-3B	Sycophancy	0.675	**0.675**	+0.1	**0.675**	-2.7	**0.675**	-0.7
	Gender	0.539	0.857	-1.3	0.783	-0.1	**0.548**	-4.1
	Safety	0.510	0.718	+0.5	0.520	+0.5	**0.510**	+0.0
Gemma-2-2B	Sycophancy	0.543	0.643	+0.0	**0.543**	-0.4	**0.543**	-0.2
	Gender	0.539	0.722	+0.3	0.681	-0.7	**0.539**	-4.9
	Safety	0.510	0.715	+0.7	0.654	+0.7	**0.510**	+0.0
Gemma-2-9B	Sycophancy	0.569	0.751	-0.1	**0.569**	-0.7	0.569	+0.0
	Gender	0.539	0.638	+0.2	0.562	-0.4	**0.539**	-3.5
	Safety	0.510	0.623	+0.0	**0.510**	+0.0	**0.510**	+0.0
Gemma-2-27B	Sycophancy	0.553	0.695	-0.2	**0.553**	-0.9	**0.553**	-0.3
	Gender	0.539	0.663	+0.0	0.605	-2.5	**0.539**	-0.5
	Safety	0.510	0.736	+0.7	0.643	+1.4	**0.510**	+0.0
Llama-3.2-1B	Sycophancy	0.551	0.849	-0.1	**0.551**	-0.2	**0.551**	+0.0
	Gender	0.539	0.855	-0.4	0.738	+0.8	**0.536**	-3.9
	Safety	0.510	0.817	+0.3	0.778	+0.3	**0.498**	+0.0
Llama-3.2-3B	Sycophancy	0.551	0.896	-0.0	**0.551**	+0.0	**0.551**	+0.0
	Gender	0.539	0.889	+0.2	0.794	-0.4	**0.535**	-2.8
	Safety	0.510	0.837	+0.1	0.821	+0.2	**0.526**	+0.0
Mistral-7B	Sycophancy	0.551	0.920	+0.0	**0.551**	-0.1	**0.551**	+0.2
	Gender	0.539	0.851	-0.4	0.815	-0.3	**0.606**	-4.8
	Safety	0.510	0.825	+0.0	0.811	+0.0	**0.532**	+0.2
Gemma-3-1B	Sycophancy	0.551	0.727	-0.1	**0.551**	-2.2	**0.551**	-0.0
	Gender	0.539	0.708	+0.0	0.613	-0.8	**0.539**	+2.0
	Safety	0.510	0.726	+0.0	0.723	-1.0	**0.510**	+1.0
Gemma-3-4B	Sycophancy	0.551	**0.551**	-0.0	**0.551**	-0.9	**0.551**	-0.4
	Gender	0.539	0.704	-0.5	0.580	-2.7	**0.539**	+4.9
	Safety	0.510	0.735	+1.0	0.609	+1.0	**0.510**	-0.2
Gemma-3-12B	Sycophancy	0.556	0.613	+0.0	**0.556**	-0.5	**0.556**	-0.5
	Gender	0.539	**0.539**	+3.5	0.606	-8.7	0.539	+3.1
	Safety	0.510	0.589	+3.8	**0.510**	+3.8	**0.510**	+0.0
Gemma-3-27B	Sycophancy	0.526	0.586	-0.2	0.566	-0.1	**0.526**	+0.0
	Gender	0.539	0.700	+1.7	0.678	-2.6	**0.539**	-4.8
	Safety	0.510	0.686	+0.9	**0.510**	+0.7	**0.510**	+0.0
Mean sycophancy		0.560	0.693	-0.0	0.563	-0.9	**0.560**	-0.1
Mean gender		0.539	0.743	+0.2	0.687	-1.6	**0.549**	-2.1
Mean safety		0.510	0.732	+0.8	0.625	+0.8	**0.512**	+0.1

Table 4: **Per-panel breakdown.** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-panel majority-class floor, subject to $|\Delta Y| \leq B$pp control-damage budget. Bold = best (smallest absolute deviation from floor) per panel per budget block. '–' = no in-budget step at that budget for that method. Final rows: per-budget per-method mean across the 39 panels and TanGCE W/T/L counts vs. the best other method (0.5pp tie band).

Panel	$\Delta Y \leq 1$pp							$\Delta Y \leq 3$pp							$\Delta Y \leq 5$pp							$\Delta Y \leq 10$pp						
	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan
Gemma2-27B/Gen L23	+12.3	–	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**
Gemma2-27B/Saf L23	+22.5	+13.3	+28.6	+22.3	-0.7	-1.7	**-0.1**	+22.5	+13.3	+28.6	+22.3	-0.7	-1.0	**-0.1**	+22.5	+13.3	+28.6	+22.3	-0.7	**-0.1**	-0.1	+22.5	+13.3	+28.6	+22.3	-0.7	**-0.1**	-0.1
Gemma2-27B/Syc L23	+14.3	**+0.0**	+10.7	+14.8	+8.5	+18.7	+0.5	+14.3	**+0.0**	+10.7	+14.8	+0.5	+18.7	+0.5	+14.3	+0.0	+10.7	+14.8	+0.5	+15.5	**-0.0**	+14.3	+0.0	+10.7	+14.8	+0.5	+15.5	**-0.0**
Gemma2-2B/Gen L13	+18.3	+14.2	+15.1	**+5.6**	+24.0	+18.5	+6.0	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+3.4**	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+0.0**	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+0.0**
Gemma2-2B/Saf L13	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**-0.0**	+2.8
Gemma2-2B/Syc L13	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.9	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.2	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.2	+9.9	**+0.0**	+2.5	+2.8	+1.6	+12.9	+0.2
Gemma2-9B/Gen L21	+9.9	+2.3	+5.1	+18.6	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+4.8	+15.4	**+0.0**
Gemma2-9B/Saf L21	+11.3	**+0.0**	+16.5	+7.2	-0.4	+2.0	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0
Gemma2-9B/Syc L21	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+1.0	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+0.2	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+0.2	+18.2	**-0.0**	+7.5	+10.6	+1.4	+20.9	+0.2
Gemma3-12B/Gen L24	-0.0	–	–	–	**+0.0**	+8.8	**+0.0**	-0.0	–	+11.0	–	**+0.0**	-0.9	**+0.0**	-0.0	–	+10.3	–	**+0.0**	**+0.0**	**+0.0**	-0.0	+6.7	+6.3	–	**+0.0**	**+0.0**	**+0.0**
Gemma3-12B/Saf L24	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1
Gemma3-12B/Syc L24	+5.7	**+0.0**	+0.8	+4.4	+1.1	–	+0.4	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.1	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.0	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.0
Gemma3-1B/Gen L13	+16.9	+7.4	+8.5	–	+5.0	–	**+0.0**	+16.9	+7.4	+2.1	+15.0	+5.0	+15.4	**+0.0**	+16.9	+7.4	**+0.0**	+12.7	+5.0	+15.4	**+0.0**	+16.9	+7.4	**+0.0**	+12.7	**+0.0**	+15.4	**+0.0**
Gemma3-1B/Saf L13	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.3	**+0.1**
Gemma3-1B/Syc L13	+17.6	–	+1.0	+4.9	+0.3	+0.3	**+0.2**	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1
Gemma3-27B/Gen L31	+16.0	–	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+24.4	+23.9	**+0.0**	–	**+0.0**
Gemma3-27B/Saf L31	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2
Gemma3-27B/Syc L31	+6.0	+4.0	**+0.3**	-0.3	+1.8	+4.3	**+0.3**	+6.0	+4.0	+0.3	-0.3	+1.8	+4.3	**+0.2**	+6.0	+4.0	+0.3	-0.3	+1.8	+4.3	**+0.2**	+6.0	+4.0	+0.3	-0.3	+1.8	+3.5	**+0.2**
Gemma3-4B/Gen L17	+16.5	–	+8.5	+19.5	+7.9	–	**+0.0**	+16.5	+4.0	+8.5	+19.5	+7.9	+15.0	**+0.0**	+16.5	+4.0	+8.5	+19.5	+7.9	+15.0	**+0.0**	+16.5	+4.0	+8.5	+19.5	**+0.0**	+15.0	**+0.0**
Gemma3-4B/Saf L17	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**
Gemma3-4B/Syc L17	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	–	+0.3	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	+0.3	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	-0.0	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	-0.0
Llama3.2-1B/Gen L8	+31.6	+19.9	+21.5	**+4.6**	+15.4	–	+16.2	+31.6	+19.9	+21.5	**+2.3**	+15.4	–	+5.8	+31.6	+19.9	+21.5	+2.3	+15.4	+25.7	**+0.0**	+31.6	+19.9	+21.5	+2.3	+15.4	+25.7	**+0.0**
Llama3.2-1B/Saf L8	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-1.7	**+0.1**
Llama3.2-1B/Syc L8	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0
Llama3.2-3B/Gen L14	+35.0	+25.5	+31.7	**+2.5**	+15.8	–	+15.1	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**-0.1**	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**-0.1**	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**+0.0**
Llama3.2-3B/Saf L14	+32.7	+31.0	+33.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**
Llama3.2-3B/Syc L14	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	+0.3	+0.3
Mistral-7B/Gen L16	+31.2	+27.6	+29.5	**+8.1**	+24.6	–	+19.5	+31.2	+27.6	+29.5	**+8.1**	+24.6	+28.6	+10.6	+31.2	+27.6	+29.5	+8.1	+24.6	+28.6	**+0.3**	+31.2	+27.6	+29.5	+8.1	+24.6	+28.6	**+0.0**
Mistral-7B/Saf L16	+31.5	+30.0	+34.6	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1
Mistral-7B/Syc L16	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.9	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3
Qwen2.5-0.5B/Gen L12	+17.0	+14.5	+13.1	**+5.4**	+6.6	+15.8	+8.8	+17.0	+14.5	+13.1	**+5.4**	+6.6	+15.8	+5.9	+17.0	+14.5	+13.1	+5.4	+6.6	+15.8	**+3.0**	+17.0	+14.5	+13.1	+5.4	**+0.0**	+15.8	+1.8
Qwen2.5-0.5B/Saf L12	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**
Qwen2.5-0.5B/Syc L12	**+0.0**	–	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2
Qwen2.5-1.5B/Gen L14	+28.2	–	+22.3	**+4.1**	+13.5	+28.2	+13.5	+28.2	+24.9	+22.3	**+4.1**	+8.9	+28.2	+11.1	+28.2	+24.9	+22.3	**+4.1**	+8.9	+28.2	+6.8	+28.2	+24.9	+22.3	+4.1	+8.9	+28.2	**+3.9**
Qwen2.5-1.5B/Saf L14	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.7	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0
Qwen2.5-1.5B/Syc L14	**+0.0**	–	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3
Qwen2.5-3B/Gen L18	–	+24.4	+24.8	**+4.7**	+14.3	–	+12.6	+31.8	+24.4	+24.8	**+4.7**	+14.3	+26.5	+8.1	+31.8	+24.4	+24.8	**+4.7**	+14.3	+26.5	+7.1	+31.8	+24.4	+24.8	+4.7	+14.3	+26.5	**+3.8**
Qwen2.5-3B/Saf L18	+20.8	+0.9	+22.3	+15.4	-0.3	-0.2	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**
Qwen2.5-3B/Syc L18	**+0.0**	–	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6
Mean (pp)	+18.3	+9.6	+14.1	+9.5	+4.3	+7.3	+2.6	+18.6	+8.9	+13.8	+9.4	+4.0	+9.3	+1.3	+18.6	+8.9	+13.7	+9.4	+4.0	+9.7	+0.6	+18.6	+8.9	+13.6	+9.4	+2.7	+9.5	+0.4
TanGCE W/T/L	7/22/10							9/23/7							10/25/4							8/28/3						

Table 5: **CelebA per-attribute breakdown (40 attributes $\times$ 2 control regimes).** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -5$pp on the 5 controls. Bold = best (smallest absolute deviation from floor) per attribute per regime block. '–' = no in-budget step at that budget for that method. Final rows: per-regime per-method mean and TanGCE W/T/L counts vs. the best other method (0.5pp tie band).

Concept	5 least-correlated controls							5 most-correlated controls						
	LE	L+C	INLP	IGBP	Obl	Amb	Tan	LE	L+C	INLP	IGBP	Obl	Amb	Tan
5_o_Clock_Shadow	+3.0	+1.3	**+0.0**	+3.0	**+0.0**	**+0.0**	**+0.0**	+3.0	+1.3	**+0.0**	+3.0	–	+1.3	**+0.0**
Arched_Eyebrows	+9.7	+3.0	+6.7	+6.7	**+0.0**	-1.7	**+0.0**	+9.7	+3.0	+6.7	+6.7	–	+4.7	**+0.0**
Attractive	+15.3	+17.3	+17.7	+1.0	-12.0	+15.7	**-0.3**	+15.3	+17.3	+17.0	**+1.0**	–	+16.0	+13.7
Bags_Under_Eyes	+0.3	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**
Bald	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Bangs	+4.7	**+0.0**	**+0.0**	+3.7	**+0.0**	+1.7	**+0.0**	+4.7	**+0.0**	**+0.0**	+3.7	**+0.0**	+1.7	**+0.0**
Big_Lips	+1.7	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+1.7	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.7	**+0.0**
Big_Nose	+2.3	+2.3	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**	+2.3	+2.3	**+0.0**	+2.0	**+0.0**	**+0.0**	-0.3
Black_Hair	+10.7	+7.7	+8.3	+4.0	**+0.0**	+6.7	+0.7	+10.7	+7.7	+8.3	+4.0	**+0.0**	**+0.0**	+1.0
Blond_Hair	+5.0	+3.0	+3.0	+3.3	**+0.0**	**+0.0**	**+0.0**	+5.0	+3.0	+3.0	+3.3	**+0.0**	+5.3	**+0.0**
Blurry	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**	**-0.0**
Brown_Hair	+0.7	-3.7	**+0.0**	+0.7	**+0.0**	-0.3	**+0.0**	+0.7	-3.7	**+0.0**	+0.7	**+0.0**	–	**+0.0**
Bushy_Eyebrows	+1.0	-0.7	+0.7	+3.7	**-0.0**	-0.3	**-0.0**	+1.0	-0.7	+0.7	+3.7	–	–	**-0.0**
Chubby	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	+0.7	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**
Double_Chin	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Eyeglasses	+2.7	**+0.0**	**+0.0**	+6.0	**+0.0**	**+0.0**	**+0.0**	+2.7	**+0.0**	**+0.0**	+6.0	**+0.0**	–	**+0.0**
Goatee	**+0.0**	**+0.0**	**+0.0**	-0.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	-0.3	**+0.0**	–	**+0.0**
Gray_Hair	**+0.0**	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**
Heavy_Makeup	+25.0	+26.0	+26.0	+23.7	**+0.0**	**+0.0**	**+0.0**	+25.0	+26.0	+26.0	+23.7	–	+25.3	**+17.7**
High_Cheekbones	+21.7	+18.7	+21.3	+14.7	**-0.0**	**-0.0**	-0.3	+21.7	–	+21.3	**+14.7**	–	+17.0	–
Male	+37.3	+37.7	+36.3	+33.7	**+0.0**	**+0.0**	-0.3	+37.3	+37.7	+36.3	**+33.0**	–	+37.3	+33.7
Mouth_Slightly_Open	+13.7	+11.7	+14.0	+5.3	**+0.0**	+0.7	**+0.0**	+13.7	+11.7	+14.0	+5.3	**+0.0**	+12.7	**+0.0**
Mustache	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	–	**+0.0**
Narrow_Eyes	**+0.0**	**+0.0**	**+0.0**	**+0.0**	–	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
No_Beard	+7.0	+1.7	+5.7	+8.0	**+0.0**	**+0.0**	**+0.0**	+7.0	+1.7	+5.7	+8.0	**+0.0**	+2.3	**+0.0**
Oval_Face	+0.3	**-0.0**	**-0.0**	**-0.0**	**-0.0**	-1.7	**-0.0**	+0.3	**-0.0**	**-0.0**	**-0.0**	**-0.0**	-4.0	**-0.0**
Pale_Skin	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Pointy_Nose	-5.0	-7.3	-0.3	+2.0	**+0.0**	**+0.0**	**+0.0**	-5.0	-7.3	-0.3	+2.0	**+0.0**	-2.0	**+0.0**
Receding_Hairline	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	**+0.0**
Rosy_Cheeks	+0.3	**-0.0**	**-0.0**	+1.3	**-0.0**	**-0.0**	**-0.0**	+0.3	**-0.0**	**-0.0**	+1.3	**-0.0**	–	**-0.0**
Sideburns	+0.3	**+0.0**	**+0.0**	+0.7	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	**+0.0**	+0.7	**+0.0**	-0.3	**+0.0**
Smiling	+30.3	+26.7	+22.3	+16.0	**+0.0**	-0.3	**+0.0**	+30.3	+26.7	+22.3	**+17.3**	–	+25.3	–
Straight_Hair	**+0.0**	+0.3	**+0.0**	+2.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+0.3	**+0.0**	+2.3	–	+0.3	**+0.0**
Wavy_Hair	+14.0	+10.7	+8.0	+15.3	**-0.0**	**-0.0**	**-0.0**	+14.0	+10.7	+8.0	+15.3	**-0.0**	–	-0.3
Wearing_Earrings	+2.7	**+0.0**	+0.3	+6.7	**+0.0**	**+0.0**	**+0.0**	+2.7	**+0.0**	+0.3	+6.7	**+0.0**	**+0.0**	**+0.0**
Wearing_Hat	**+0.0**	**+0.0**	**+0.0**	+2.3	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	+2.3	**+0.0**	**+0.0**	**+0.0**
Wearing_Lipstick	+40.0	+38.0	+39.0	+27.7	-4.3	**+0.0**	+0.3	+40.0	+38.0	+39.0	**+30.0**	–	+38.3	+33.0
Wearing_Necklace	-1.0	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	-1.0	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**	**+0.0**
Wearing_Necktie	+1.3	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**	+1.3	**+0.0**	**+0.0**	+2.0	**+0.0**	**+0.0**	**+0.0**
Young	+11.0	+9.3	+11.0	+12.3	**-0.0**	+8.7	**-0.0**	+11.0	+9.3	+11.0	+12.3	**-0.0**	+8.7	+0.3
Mean (pp)	+6.4	+5.1	+5.5	+5.3	-0.4	+0.7	+0.0	+6.4	+4.7	+5.5	+5.3	+0.0	+5.8	+2.6
TanGCE W/T/L				1/38/1							3/31/4			

Table 6: **Per-row λ_i statistics for the proposed pipeline ($\varepsilon = 0.1$, $\lambda_{\max} = 64$) on the 39-panel NLP grid at 50%-depth.** Each cell averages the per-row λ_i first across all 60 outer iterations of one trajectory, then across the 13 model panels of the concept. The within-panel p_{10}/p_{90} spread (mean across panels) is roughly $14/45$, i.e. a $\sim$10$\times$ ratio between the least- and most-confident rows in a single iteration. The safety cap $\lambda_{\max} = 64$ is observed only sporadically (max 57.32 across 39$\times$60 = 2340 iterations); the closed-form rule rarely binds against it.

Concept	n	mean λ_i	median	min	max
Sycophancy	13	29.58	37.15	4.89	49.38
Gender	13	25.28	23.76	0.45	45.08
Safety	13	33.07	33.30	1.71	57.32
Overall	**39**	**29.31**	**30.19**	**0.45**	**57.32**